

1

Review Mode ⓘ

Every year, the city of Whoville conducts a Reindeer Run to find the best reindeers for Santa's Sleigh. Can you write a query to return the name and rank of the top 7 reindeers in this race?

Tables Explore data

reindeer_run_results(number, name, rank, color)

- ① Select name, rank
- ② Using limit 7 cuz limit is filter only first row until number row you want.
- ③ Order by rank from MAX → MIN via DESC keywords.

2

Review Mode ⓘ

Santa wants to analyze which toys that were produced in his workshop have already been delivered to children. You are given two tables on toy production and toy delivery — can you return the toy ID and toy name of the toys that have been delivered?

Tables Explore data

toy_production(toy_id, toy_name, production_date)

toy_delivery(toy_id, child_name, delivery_date)

- wanna know which toys that produced and delivered to children.
- want 2 column (toy_id, toy_name)

- ① Choosing toy_id, name in SELECT
 - ② JOIN 2 table together with this purpose is only row have data to connect so we can cut value in Deliverate is NULL, so we choose INNER JOIN
- by use column toy_id to be condition to connect.

3

Review Mode ①

The Grinch has brainstormed a ton of pranks for Whoville, but he only wants to keep the top prank per target, with the highest evilness score. Return the most evil prank for each target. If two pranks have the same evilness, the more recently brainstormed wins.

Tables [Explore data](#)

`grinch_prank_ideas(prank_id, target_name, prank_description, evilness_score, created_at)`

- ① This question wanna know what prank per target with highest evilness_score
- ② We need window-function to order rank via evilness_score from MAX to MIN (RANK cuz they will count in each group as column in Partition By) and if same evilness_score, they will compare most recently brainstormed wins (created_at DESC)
- ③ We will use previous step as temporary table and query from this cuz in WHERE statement can use only name of column and can't use any aggregate function.

4

Review Mode ①

Kevin's trying to decorate the house without sending the electricity bill through the roof. Write a query to find the top 5 most energy-efficient decorations (i.e. lowest cost per hour to operate).

Tables [Explore data](#)

`hall_decorations(decoration_id, decoration_name, energy_cost_per_hour)`

This question wanna know most top 5 use electric lowest cost

- ① Using column about cost per hour and ORDER BY cost per hour to sort min→max
- ② Using LIMIT 5 for filter only five rows first to show Output.

5

Review Mode ①

Some elves took time off after the holiday rush, but not everyone has returned to work. List all elves by name, showing their return date. If they have not returned from vacation, list their return date as "Still resting". ^

Tables

Explore data

elves(elf_id, elf_name)

vacations(elf_id, start_date, return_date)

This question wanna know when that Elves will return and they name, and if that returned_date is empty will replace with "still resting"

- ① Choosing elf_name, return_date in SELECT
- ② JOIN 2 table with LEFT JOIN cuz some column in second if empty they still combine by replace with NULL, so you can make every row can join even empty.
- ③ Using COALESCE to handle NULL value by replace with "still resting"

COALESCE(return_date, 'still resting')

6

Review Mode ①

Buddy is planning a winter getaway and wants to rank ski resorts by annual snowfall. Can you help him bucket these ski resorts into quartiles? ^

Tables

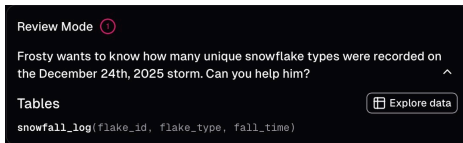
Explore data

resort_monthly_snowfall(resort_id, resort_name, snow_month, snowfall_inches)

This question wanna know about what resort is perfect for booking by ranking all into quartiles.

- ① Sum snowfall_inches each resort and GROUP BY resort_name with temporary table
- ② Using resort_name, sum-of-snowfall_inches, quartiles in SELECT.
- ③ Ranking that resource via NTILE(4) and sort min→max with sum-of-snowfall_inches.

7



This question wanna know how many unique snowflake type as recorded in 2025-12-24

① Using **flake_type** in SELECT with **DISTINCT** (Prevent Duplicate Value) and **COUNT**

SELECT COUNT(DISTINCT flake_type)

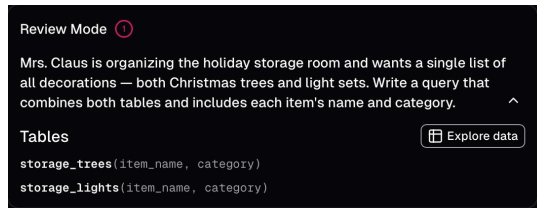
② Filter **fall_time** only **2025-12-24** in WHERE (but fall_time datatype not DATE but TIMESTAMP so we need to cast to DATE)

WHERE fall_time::DATE = '2025-12-24'

③ Group by with **flake_type**

GROUP BY flake_type

8



This question wanna organize data from 2 tables in one table to easy to see data

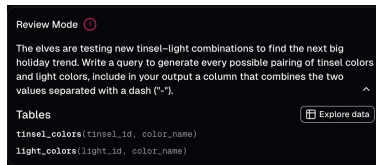
① We need to combine 2 tables → 1 table in single list (Vertical)

so we choose to use **UNION**.

JOIN vs UNION

JOIN	connect 2 table in 1 table in horizontal
UNION	connect 2 table in 1 table in vertical.

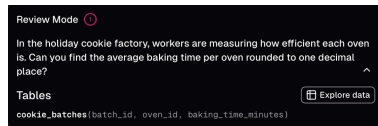
9



This question want output show every possible match color between tinsel & light $\rightarrow$ Cartesian product ($A \times B$)

- ① Using CROSS JOIN cuz every row from first table is combined with every table from second tables
- ② Using Concat function to combine two value with "-"
like dark-white, etc... $\rightarrow$ CONCAT(value1, value2, value3, ...)
by value can replace with variable from column name, static string

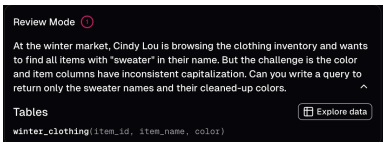
10



This question wanna know avg() baking time per oven and rounded decimal $\rightarrow$ 1 decimal cuz batch_id is number of test, oven_id is number of oven so 1 oven have test many time, and make differece time output.

- ① Choosing baking-time-minutes with SUM(baking-time-minutes) for sum all each oven, after that will use AVG() for find average per oven and that will use round in 1 decimal, and oven-id cuz output wanna see in each oven.
- ② When you use aggregation function, you need to group your output by GROUP-BY oven-id

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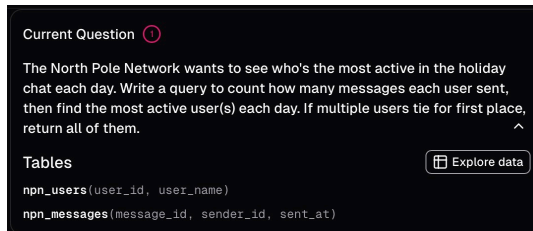
This question wanna know what clothing name include "sweater" but in table a item-name is inconsistent capital so do color too.

- ① We need to cleaned-up name in LOWER case cuz a word that they want to find in each row item-name is LOWER and use wildcard like `%`, `_` by `%` and `_` is show how many can be in that place by `%` is zero or more
`_` is zero or one

`LOWER(item_name) LIKE '%sweater%'` cuz `%` is zero or more word can replace.

- ② Choosing item_name, color in SELECT and cleaned-up to LOWER.

`item_name, LOWER(color)`



This question wanna who sends message most in each day

- We need count message as sent each person via sent.id in CTE and cast `TIMESTAMP` to `DATE`
 - `COUNT(message_id)` AS frequency
 - `sent_at::DATE` AS time-to-sent
- We use `WINDOW` function for contain same row with last CTE via `RANK` cuz when same value each row rank number will same but difference from `ROW_NUMBER` just `1 → ∞` by increase 1 after pass 1 row.
 - `RANK() OVER(PARTITION BY time-to-sent ORDER BY frequency DESC)`
- We will use last CTE to join with `nnp-users` for ger name from sender_id cuz `sender_id = user_id` with `LEFT JOIN` or `INNER JOIN` cuz want output show name follow row in last CTE but if you not sure, can use `INNER JOIN` but if sure every row each table have data, you can use `LEFT`

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13

Review Mode ①

Santa's audit team is reviewing this year's behavior scores to find the extremes — write a query to return the lowest and highest scores recorded on the Naughty or Nice list. ^

Tables Explore data

`behavior_scores(record_id, child_name, behavior_score)`

This question wanna lowest/highest behavior scores recorded on list

- ① Using `MIN()` to find lowest, and `MAX()` to find highest with `behavior_score` column.
- `SELECT MIN(behavior_score), MAX(behavior_score)`

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Current Question ①

The Productivity Club is tracking members' challenge start dates and wants to calculate each member's `focus_end_date`, exactly 14 days after their start date. Can you write a query to return the existing table with the `focus_end_date` column? ^

Tables Explore data

`focus_challenges(member_id, member_name, start_date)`

This question wanna know about end-date each member cuz they didn't start in the same day but want all column + `focus_end_date` in output.

- Find end date via use `start_date + 14` (Only PostgreSQL) alias with `focus_end_date`
- Using `*` with `focus_end_date`
`SELECT *, focus_end_date`

15

Current Question 1

The Grinch is tracking his daily mischief scores to see how his behavior changes over time. Can you find how many points his score increased or decreased each day compared to the previous day?

Tables

Explore data

grinch_mischief_log(log_date, mischief_score)

This question wanna know how many scores point difference from yesterday

- Use LAG() in window function to get previous rows data (yesterday data)
- We find how many point change each day via $\text{mischief_score} - \text{LAG}(\text{mischief_score})$
- Output have 2+2 column in Output



Here's my SQL query for question 1:

```
SELECT *,
       LAG(mischief_score) OVER(ORDER BY
log_date) AS previous_day_scores,
       mischief_score-LAG(mischief_score)
OVER(ORDER BY log_date) AS
scores_day_changes
FROM grinch_mischief_log;
```

Optimize query

16

Review Mode 1

It's a snow day, and Buddy is deciding which tasks he can do from under a blanket. Can you find all tasks that are either marked as 'Work From Home' or 'Low Priority' so he can stay cozy and productive?

Tables

Explore data

daily_tasks(task_id, task_name, task_type, priority)

This question wanna know which tasks he can do in home and keyword is type is WFH or Priority is LOW

- filter in WHERE clause with 2 column first (task-type), Second (priority)



Here's my SQL query for question 1:

```
SELECT *
FROM daily_tasks
WHERE priority = 'Low' OR
task_type = 'Work From Home';
```

Optimize query

17

Current Question ①

During a quiet evening of reflection, Cindy Lou wants to categorize her tasks based on how peaceful they are. Can you write a query that adds a new column classifying each task as 'Calm' if its noise_level is below 50, and 'Chaotic' otherwise?

Tables

evening_tasks(task_id, task_name, noise_level)

Explore data

This question wanna categorize tasks on how peaceful with new column in output if < 50 show calm, otherwise Chaotic

- ① Using CASE WHEN for show output dynamic with *



Here's my SQL query for question 1:

```
SELECT *,
CASE
WHEN noise_level < 50 THEN 'Calm'
ELSE 'Chaotic'
END AS peaceful
FROM evening_tasks;
```

Optimize query

*** New knowledge

18

Review Mode ①

Over the 12 days of her data challenge, Data Dawn tracked her daily quiz scores across different subjects. Can you find each subject's first and last recorded score to see how much she improved?

Tables

daily_quiz_scores(subject, quiz_date, score)

Explore data

This question wanna know first scores & last scores each subject and show only 1 rows each subject.

- ① find first & last scores with FIRST_VALUE() & LAST_VALUE() with window-function and group each subject
- ② After use window function they will keep row output same with table that's mean they have many duplicate row, so we use ROWS BETWEEN UNBOUNDED PRECEDING AND UNBOUNDED FOLLOWING



Here's my SQL query for question 1:

```
SELECT DISTINCT *
FROM (
SELECT subject,
FIRST_VALUE(score) OVER(
PARTITION BY subject
ORDER BY quiz_date) AS
first_values,
LAST_VALUE(score) OVER(
PARTITION BY subject
ORDER BY quiz_date
ROWS BETWEEN UNBOUNDED PRECEDING
AND UNBOUNDED FOLLOWING
) AS last_values
FROM daily_quiz_scores) AS
improve_scores;
```

Optimize query

WINDOW FRAME

Frame types : ROWS, RANGE

Frame boundaries

→ lower value :

- CURRENT ROW
- N PRECEDING → number of row to start
- UNBOUNDED PRECEDING : first row

→ higher value :

- CURRENT ROW
- N FOLLOWING → number of row to end.
- UNBOUNDED FOLLOWING → last row

* Lower value much less than higher value

Ex. ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW

19

Current Question 1

Clara is reviewing holiday orders to uncover hidden patterns — can you return the total amount of wrapping paper used for orders that were both gift-wrapped and successfully delivered?

Tables

Explore data

holiday_orders(order_id, customer_name, gift_wrap, paper_used_meters, delivery_status, order_date)

This question wanna know how many wrapping paper used for order gift-wrapped & successfully delivered.

- ① Using WHERE to filter only column gift-wrapped, successfully delivered
- ② Use sum function to find how many wrapping paper used?

Here's my SQL query for question 1:

```
SELECT SUM(paper_used_meters)
FROM holiday_orders
WHERE (gift_wrap = TRUE) AND
      (delivery_status = 'Delivered');
```

Optimize query

20

Current Question 1

Jack Frost wants to review all the cocoa breaks he actually took — including the cocoa type and the location he drank it in. How would you combine the necessary tables to show each logged break with its matching cocoa details and location?

Tables

Explore data

cocoa_logs(log_id, break_id, cocoa_id)

break_schedule(break_id, location_id)

cocoa_types(cocoa_id, cocoa_name)

locations(location_id, location_name)

This question wanna know each break about cocoa with cocoa type & location

- ① JOIN table together for query break_id, cocoa_id, cocoa_name, location_name

Here's my SQL query for question 1:

```
SELECT cocoa_logs.break_id,
       cocoa_types.cocoa_id,
       cocoa_types.cocoa_name,
       locations.location_name
FROM cocoa_logs
JOIN break_schedule ON
  break_schedule.break_id =
  cocoa_logs.break_id
JOIN cocoa_types ON
  cocoa_types.cocoa_id =
  cocoa_logs.cocoa_id
JOIN locations ON
  locations.location_id =
  break_schedule.location_id
```

Optimize query

21

Current Question 1

The Snow Queen hosts nightly fireside chats and records how many stories she tells each evening. Can you calculate the running total of stories she has shared over time, in the order they were told?

Tables

story_log(log_date, stories_shared)

Explore data

This question wanna know how many stories as she said in each evening, that's day by day

- ① We need to use window function by SUM cuz this function can combine value and use with window frame for combine value each column

ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW

↓
row by row

↓
start from first row

↓
current row as statement execute



Here's my SQL query for question 1:

```
SELECT *,
       SUM(stories_shared) OVER(
         ORDER BY log_date
         ROWS BETWEEN UNBOUNDED PRECEDING
         AND CURRENT ROW
       ) AS total_stories
FROM story_log
```

Optimize query

22

Review Mode 1

The penguins are signing up for a community sleigh ride, but the organizers need a list of everyone who did NOT choose the "Evening Ride." How would you return all penguins whose selected time is not the evening slot?

Tables

sleigh_ride_signups(signup_id, penguin_name, ride_time)

Explore data

This question wanna know which penguins signed up for join community sleigh ride not ride-time in evening

- ① Using penguin_name for show output
② filter ride-time ≠ 'evening'



```
SELECT penguin_name
FROM sleigh_ride_signups
WHERE ride_time ≠ 'Evening';
```

23

Review Mode 1

The Gingerbread House Competition wants to feature the builders who used at least 4 distinct candy types in their designs. How would you identify these builders?

Tables

Explore data

gingerbread_designs(builder_id, builder_name, candy_type)

This question wanna know what builder_name in this competition use candy at least 4 by not same types.

- ① Count candy-type each builder-name with DISTINCT
- ② Using Having to filter ① only $\geq$ (greater than or equal)

4



Here's my SQL query for question 1:

```
SELECT builder_name
FROM (
  SELECT builder_name,
    COUNT(DISTINCT candy_type) AS
    candy_type_distinct
  FROM gingerbread_designs
  GROUP BY builder_name
  HAVING COUNT(DISTINCT candy_type) >=
  4) AS builder_name_with_candy_four;
```

Optimize query

24

Current Question 1

As the New Year begins, the goals tracker team wants to understand how user types differ. How many completed goals does the average user have in each user_type?

Tables

Explore data

user_goals(user_id, user_type, goal_id, goal_status)

This question wanna know how many completed goals as average user in each user_type

- ① We need to filter only goal-status have Completed
- ② We need to group that data with COUNT and Group with user_id and created with CTE, so now we already have table as contain how many goal are completed per users.
- ③ We find an average completed each user-type cuz if we count with user_type after that they can't find avg so we need to group with user.id before used avg each type.

Calendar
Here's my SQL query for question 1:

```
WITH
  number_of_complete_each_users
AS (
  SELECT user_id,
    user_type,
    COUNT(goal_status) AS
    number_of_complete
  FROM user_goals
  WHERE goal_status =
    'Completed'
  GROUP BY user_id, user_type
)
SELECT user_type,
  AVG(number_of_complete)
FROM
  number_of_complete_each_users
GROUP BY user_type;
```

Optimize query